AJ_24038155_dialect_v1.1

Full integrated Cambridge-Style Pseudocode specification

Integrated specification: This document incorporates the complete v1.0 dialect and the approved v1.1 language additions into a single normative specification.

1. Language Overview

The AJ_24038155 Dialect is a structured pseudocode language intended for:

- representing algorithms generated from natural-language tasks;
- deterministic parsing into an Abstract Syntax Tree;
- conversion into a Control Flow Graph;
- visual generation of flow diagrams;
- deterministic translation into Python;
- safe behavioural evaluation.

The language uses explicit block terminators and does not rely on indentation for program structure.

Keywords are case-insensitive during parsing but are presented in uppercase as the canonical format.

Identifiers remain case-sensitive.

2. Lexical Rules

2.1 Identifiers

An identifier must begin with a letter or underscore.

```
identifier ::= letter , { letter | digit | "_" } ;
```

Examples:

total

item_count

_grid

value2

Invalid examples:

2value

item-count

total value

2.2 Integer Literals

```
integer_literal ::= [ "-" ] , digit , { digit } ;
```

Examples:

0

12

-5

1000

2.3 Float Literals

```
float_literal ::= [ "-" ] , digit , { digit } , "." , digit , { digit } ;
```

Examples:

0.5

3.14

-10.25

2.4 String Literals

Strings use the STRING constructor with quoted contents.

```
string_literal ::= "STRING" , "(" , quoted_string , ")" ;
```

Examples:

```
STRING("Hello")
STRING("Enter a number")
STRING("The result is {result}")
```

Escape sequences may be used:

\"

\\

\n

\t

Example:

```
STRING("He said \"Hello\"")
```

2.5 Character Literals

Characters use the CHAR constructor and must contain exactly one character or one valid escape sequence.

```
char_literal ::= "CHAR" , "(" , quoted_character , ")" ;
```

Examples:

```
CHAR("a")
CHAR("7")
CHAR("\n")
```

2.6 Boolean Literals

```
boolean_literal ::= "BOOLEAN" , "(" , ( "TRUE" | "FALSE" ) , ")" ;
```

Canonical examples:

```
BOOLEAN(TRUE)
BOOLEAN(FALSE)
```

2.7 Native Null Literal

NONE is the native null literal in v1.1.

```
none_literal ::= "NONE" ;
```

Keywords are case-insensitive during parsing, therefore NONE, None and none denote the same literal.

Canonical output uses uppercase NONE.

Python translation:

```
NONE
```

translates deterministically to:

None

NONE is a value. It is not an identifier and is not a function call.

The historical compatibility form CALL NULL() remains accepted for existing v1.0 material. New canonical v1.1 pseudocode should use NONE.

2.8 Comments

Single-line comments begin with COMMENT.

```
comment ::= "COMMENT" , comment_text , newline ;
```

Example:

```
COMMENT Calculate the running total
```

Comments are ignored during AST construction but may be stored as metadata.

2.9 Whitespace and Newlines

Spaces and indentation are not semantically significant except inside string literals.

Each statement must end with a newline.

Block structure is determined by explicit keywords such as:

```
ENDIF
ENDWHILE
ENDFOR
ENDFUNCTION
```

3. Operator Rules

3.1 Assignment

Assignment uses <-.

```
variable <- value
```

Example:

```
total <- 0
name <- STRING("Alan")
```

3.2 Arithmetic Operators

+

-

*

/

```
INTEGER DIVISION
MODULUS
EXP
```

Examples:

```
sum <- a + b
difference <- a - b
product <- a * b
quotient <- a / b
whole <- a INTEGER DIVISION b
remainder <- a MODULUS b
power <- x EXP y
```

3.3 Comparison Operators

==

!=

<

>

<=

>=

Examples:

x == y

x != y

x<y

x >= y

3.4 Boolean Operators

```
NOT
```

```
AND
OR
XOR
```

Examples:

x > 0 AND x < 10

NOT finished

left OR right

flag1 XOR flag2

3.5 Operator Precedence

Operators are evaluated in the following order, from highest to lowest:

1. Parentheses
2. Function and collection access
3. EXP
4. Unary minus
5. *, /, INTEGER DIVISION, MODULUS
6. +, -
7. ==, !=, <, >, <=, >=
8. NOT
9. AND
10. XOR
11. OR

Parentheses should be used where ambiguity may occur.

4. Data Types and Literals

4.1 Supported Primitive Types

```
INTEGER
FLOAT
STRING
CHAR
BOOLEAN
```

4.2 Integer Literal

```
INTEGER(5)
```

```
INTEGER(-20)
```

Formal rule:

```
typed_integer ::= "INTEGER" , "(" , integer_literal , ")" ;
```

4.3 Float Literal

```
FLOAT(3.14)
FLOAT(-0.5)
```

Formal rule:

```
typed_float ::= "FLOAT" , "(" , float_literal , ")" ;
```

4.4 Tuple Literal

```
TUPLE(a, b)
TUPLE(INTEGER(1), STRING("A"))
```

Formal rule:

```
tuple_literal ::=
"TUPLE" ,
"(" ,
[ expression , { "," , expression } ] ,
")" ;
```

4.5 Homogeneous Typed Lists

A homogeneous list contains values of one declared type.

Examples:

```
LIST OF INTEGER [1, 2, 3]
LIST OF STRING [STRING("A"), STRING("B")]
LIST OF BOOLEAN [BOOLEAN(TRUE), BOOLEAN(FALSE)]
```

Formal rule:

```
typed_list_literal ::=
"LIST OF" ,
type_name ,
```

"[" ,

[expression , { "," , expression }] ,

"]" ;

All values must be compatible with the declared type.

4.6 Mixed Lists

Mixed lists use the LIST keyword.

Example:

```
LIST [INTEGER(5), STRING("hello"), CHAR("a"), BOOLEAN(TRUE)]
```

Formal rule:

```
mixed_list_literal ::=
```

"LIST" ,

"[" ,

[expression , { "," , expression }] ,

"]" ;

4.7 Empty Values

Supported empty values are:

```
EMPTY LIST
EMPTY DICTIONARY
EMPTY STRING
EMPTY SET
EMPTY TUPLE
```

Formal rule:

```
empty_literal ::=
```

"EMPTY LIST"

```
| "EMPTY DICTIONARY"
| "EMPTY STRING"
| "EMPTY SET"
| "EMPTY TUPLE" ;
```

4.8 Native Null Literal

NONE represents the absence of a value.

Canonical syntax:

```
NONE
```

Formal rule:

```
none_literal ::= "NONE" ;
```

Python translation:

NONE -> None

5. Program Structure

5.1 Program

A program contains one or more declarations or statements.

```
program ::= { declaration | statement | comment } ;
```

A program should normally contain at least one function when it is intended for automated testing.

5.2 Function Definition

Canonical syntax:

```
FUNCTION name ACCEPTS(parameter1, parameter2)
```
statements
```
ENDFUNCTION
```

Example:

```
FUNCTION calculate_total ACCEPTS(numbers)
total <- 0

FOR EACH number IN numbers DO
total <- total + number
ENDFOR

RETURN total
ENDFUNCTION
```

Formal rule:

```
   function_definition ::=
"FUNCTION" ,
   identifier ,
"ACCEPTS" ,
"(" ,
   [ parameter_list ] ,
")" ,
newline ,
{ statement | comment } ,
"ENDFUNCTION" ,
   newline ;
```

5.3 Parameter List

```
   parameter_list ::=
parameter ,
{ "," , parameter } ;
```

A parameter may optionally declare a type.

Examples:

```
   FUNCTION process ACCEPTS(value)
   FUNCTION process ACCEPTS(value AS INTEGER)
```

Formal rule:

```
   parameter ::=
   identifier ,
[ "AS" , type_name ] ;
```

5.4 Return Statement

```
   RETURN expression
```

or:

```
   RETURN
```

Formal rule:

```
   return_statement ::=
"RETURN" ,
[ expression ] ,
   newline ;
```

A value-less return is permitted.

5.5 Procedure and Function Calls

A call may be used as an expression:

```
   result <- CALL calculate_total(numbers)
```

or as a standalone statement:

```
   CALL display_result(result)
```

Formal rules:

```
   call_expression ::=
"CALL" ,
   identifier ,
"(" ,
   [ argument_list ] ,
")" ;

   call_statement ::=
call_expression ,
   newline ;

   argument_list ::=
   expression ,
{ "," , expression } ;
```

6. Input and Output

6.1 Input

Input must contain a prompt expression.

```
   age <- INPUT(STRING("Enter your age"))
```

Formal rule:

```
   input_expression ::=
"INPUT" ,
"(" ,
   expression ,
")" ;
```

6.2 Output

```
   OUTPUT expression
```

Examples:

```
   OUTPUT result
   OUTPUT STRING("The value is {value}")
```

Formal rule:

```
   output_statement ::=
"OUTPUT" ,
   expression ,
   newline ;
```

String placeholders use braces:

```
   STRING("The result is {result}")
```

The referenced identifier must exist in the current scope.

7. Assignment Rules

7.1 Standard Assignment

```
   identifier <- expression
```

Example:

```
   count <- 0
```

7.2 Collection Element Assignment

```
list[index] <- value
dictionary[key] <- value
```

Examples:

```
numbers[0] <- 10
visited[TUPLE(nr, nc)] <- TUPLE(r, c)
```

Formal rule:

```
assignment_statement ::=
```
assignable ,

"<-" ,

```
expression ,
newline ;
```

```
assignable ::=
```
identifier

```
| indexed_access ;
```

8. Expressions

8.1 Expression Grammar

```
expression ::= or_expression ;
```

```
or_expression ::=
```
xor_expression ,

{ "OR" , xor_expression } ;

```
xor_expression ::=
```
and_expression ,

{ "XOR" , and_expression } ;

```
and_expression ::=
```
not_expression ,

{ "AND" , not_expression } ;

```
not_expression ::=
```
"NOT" , not_expression

```
| comparison_expression ;
```

```
   comparison_expression ::=
```
additive_expression ,

[comparison_operator , additive_expression] ;

```
   additive_expression ::=
```
multiplicative_expression ,

{ ("+" | "-") , multiplicative_expression } ;

```
   multiplicative_expression ::=
```
power_expression ,

{

```
   (
```
"*"

```
   | "/"
   | "INTEGER DIVISION"
   | "MODULUS"
```
),

power_expression

};

```
   power_expression ::=
```
unary_expression ,

["EXP" , power_expression] ;

```
   unary_expression ::=
```
"-" , unary_expression

```
   | primary_expression ;
```

```
   primary_expression ::=
```
literal

```
   | identifier
   | indexed_access
   | member_call_expression
   | call_expression
   | input_expression
   | length_expression
   | join_expression
   | dictionary_get_expression
   | random_expression
   | "(" , expression , ")" ;
```

8.2 Comparison Operator

```
   comparison_operator ::=
```
"=="

```
   | "!="
   | "<"
   | ">"
   | "<="
   | ">=" ;
```

9. Conditional Rules

9.1 IF Statement

```
   IF condition THEN
```

statements

```
   ENDIF
```

Formal rule:

```
   if_statement ::=
```

"IF" ,

```
   expression ,
```

"THEN" ,

newline ,

{ statement | comment } ,

{ else_if_clause } ,

[else_clause] ,

"ENDIF" ,

```
   newline ;
```

9.2 ELSE IF Clause

```
   ELSE IF condition THEN
```

statements

Formal rule:

```
   else_if_clause ::=
```

"ELSE IF" ,

```
   expression ,
```

"THEN" ,

newline ,

{ statement | comment } ;

9.3 ELSE Clause

```
   ELSE
```

statements

Formal rule:

```
else_clause ::=
```

"ELSE" ,

newline ,

{ statement | comment } ;

9.4 Full Example

```
IF score >= 70 THEN
grade <- STRING("Distinction")
ELSE IF score >= 60 THEN
grade <- STRING("Merit")
ELSE
grade <- STRING("Pass")
ENDIF
```

10. Loop Rules

10.1 WHILE Loop

```
WHILE condition DO
```

statements

```
ENDWHILE
```

Formal rule:

```
while_statement ::=
```

"WHILE" ,

```
expression ,
```

"DO" ,

newline ,

{ statement | comment } ,

"ENDWHILE" ,

```
newline ;
```

10.2 Range Loop

Range loops are end-exclusive.

Therefore:

```
   FOR i <- IN RANGE 0 TO n DO
```

iterates through:

0, 1, 2, ..., n - 1

Canonical syntax:

```
   FOR identifier <- IN RANGE start TO end DO
```
statements
```
   ENDFOR
```

Optional step:

```
   FOR i <- IN RANGE 10 TO 0 STEP -1 DO
```
statements
```
   ENDFOR
```

Formal rule:

```
   range_for_statement ::=
```
"FOR" ,
```
   identifier ,
```
"<-" ,
"IN RANGE" ,
```
   expression ,
```
"TO" ,
```
   expression ,
```
["STEP" , expression] ,
"DO" ,
newline ,
{ statement | comment } ,
"ENDFOR" ,
```
   newline ;
```

A step value must not equal zero.

10.3 FOR EACH Loop

```
   FOR EACH item IN collection DO
```
statements

```
   ENDFOR
```

Formal rule:

```
   for_each_statement ::=
"FOR EACH" ,
   identifier ,
"IN" ,
   expression ,
"DO" ,
newline ,
{ statement | comment } ,
"ENDFOR" ,
   newline ;
```

10.4 BREAK

```
   BREAK
```

Formal rule:

```
   break_statement ::=
"BREAK" ,
   newline ;
```

BREAK is valid only inside:

```
   WHILE
   FOR
   FOR EACH
```

10.5 CONTINUE

The dialect includes an optional CONTINUE statement.

```
   CONTINUE
```

Formal rule:

```
   continue_statement ::=
"CONTINUE" ,
   newline ;
```

CONTINUE is valid only inside a loop.

11. List and Collection Rules

11.1 Indexing

collection[index]

Examples:

numbers[0]

grid[row][column]

Formal rule:

```
   indexed_access ::=
```

primary_identifier ,

{ "[" , expression , "]" } ;

```
   primary_identifier ::= identifier ;
```

Indexes begin at zero.

11.2 Length

LENGTH OF collection

Formal rule:

```
   length_expression ::=
```

"LENGTH OF" ,

expression ;

11.3 Append

```
   APPEND value TO collection
```

Formal rule:

```
   append_statement ::=
```

"APPEND" ,

```
   expression ,
```

"TO" ,
  expression ,
  newline ;

11.4 Insert

```
INSERT value INTO collection AT index
```

Formal rule:

```
  insert_statement ::=
"INSERT" ,
  expression ,
"INTO" ,
  expression ,
"AT" ,
  expression ,
  newline ;
```

11.5 Remove First Item

```
REMOVE FIRST ITEM FROM collection
```

Formal rule:

```
  remove_first_statement ::=
"REMOVE FIRST ITEM FROM" ,
  expression ,
  newline ;
```

11.6 Remove Last Item

```
REMOVE LAST ITEM FROM collection
```

Formal rule:

```
  remove_last_statement ::=
"REMOVE LAST ITEM FROM" ,
  expression ,
  newline ;
```

11.7 Remove Specific Item

```
REMOVE value FROM collection
```

Formal rule:

```
   remove_value_statement ::=
"REMOVE" ,
   expression ,
"FROM" ,
   expression ,
   newline ;
```

11.8 Join

```
   JOIN(collection, separator)
```

Example:

```
   result <- JOIN(characters, EMPTY STRING)
```

Formal rule:

```
   join_expression ::=
"JOIN" ,
"(" ,
   expression ,
"," ,
   expression ,
")" ;
```

11.9 Two-Dimensional List Creation

Canonical form:

```
   grid <- EMPTY LIST
   r <- 0

   WHILE r < rows DO
   row <- EMPTY LIST
   c <- 0

   WHILE c < cols DO
   APPEND fill_value TO row
   c <- c + 1
   ENDWHILE

   APPEND row TO grid
   r <- r + 1
   ENDWHILE
```

No special two-dimensional list syntax is required. Nested lists are constructed using ordinary list operations.

12. Set Rules

12.1 Set Literal

```
SET [value1, value2, value3]
```

Formal rule:

```
set_literal ::=
```

"SET" ,

"[" ,

[expression , { "," , expression }] ,

"]" ;

12.2 Set Operations

```
UNION(set1, set2)
INTERSECTION(set1, set2)
DIFFERENCE(set1, set2)
```

Formal rules:

```
set_operation ::=
```

"UNION" , "(" , expression , "," , expression , ")"

```
| "INTERSECTION" , "(" , expression , "," , expression , ")"
| "DIFFERENCE" , "(" , expression , "," , expression , ")" ;
```

12.3 Membership

value IN collection

value NOT IN collection

These are treated as comparison expressions.

13. Dictionary Rules

13.1 Dictionary Creation

```
visited <- EMPTY DICTIONARY
```

13.2 Dictionary Assignment

```
dictionary[key] <- value
```

Example:

```
visited[TUPLE(nr, nc)] <- TUPLE(r, c)
```

13.3 Dictionary Retrieval

Canonical v1.1 syntax:

```
value <- dictionary.GET(key)
```

Example:

```
node <- visited.GET(node)
```

Dictionary retrieval uses the general member-method invocation notation.

13.4 Dictionary Retrieval With Default

Optional syntax:

```
value <- dictionary.GET(key, default_value)
```

When resolved as the dictionary intrinsic, GET accepts exactly one or two arguments:

```
dictionary.GET(key)
dictionary.GET(key, default_value)
```

Without a default value, a missing key produces a runtime error.

13.4.1 Semantic Resolution

GET is not globally interpreted as dictionary retrieval by spelling alone.

```
The source form receiver.GET(...) is first parsed as an ordinary member-method invocation. It
resolves to the dictionary intrinsic only when static evidence establishes that receiver is
dictionary-backed, for example:
```

```
frequency <- EMPTY DICTIONARY
count <- frequency.GET(character, 0)
```

or:

```
FUNCTION lookup ACCEPTS(values AS DICTIONARY, key)
RETURN values.GET(key)
ENDFUNCTION
```

or when a class property is declared AS DICTIONARY.

If the receiver is not dictionary-backed, GET remains an ordinary user-defined method name.

Python translation:

```
dictionary.GET(key) -> dictionary.get(key)
dictionary.GET(key, default_value) -> dictionary.get(key, default_value)
```

13.4.2 Backwards Compatibility With v1.0

```
The v1.0 spellings dictionary~GET(key) and dictionary~GET(key, default_value) remain accepted
for backwards compatibility.
Under v1.1 compatibility normalisation they canonicalise to dictionary.GET(key) and
dictionary.GET(key, default_value).
```

The ~GET spelling is accepted but non-canonical in v1.1.

13.5 Dictionary Membership

key IN dictionary

key NOT IN dictionary

14. Random Rules

14.1 Random Integer

```
RANDOM INT(minimum, maximum)
```

Both endpoints are inclusive.

Formal rule:

```
random_integer_expression ::=
```

"RANDOM INT" ,

"(" ,

```
expression ,
```

"," ,

```
   expression ,
")" ;
```

14.2 Random Float

```
   RANDOM FLOAT(minimum, maximum)
```

Formal rule:

```
   random_float_expression ::=
"RANDOM FLOAT" ,
"(" ,
   expression ,
"," ,
   expression ,
")" ;
```

14.3 Random Boolean

```
   RANDOM BOOLEAN()
```

Formal rule:

```
   random_boolean_expression ::=
"RANDOM BOOLEAN" ,
"(" ,
")" ;
```

14.4 Combined Rule

```
   random_expression ::=
random_integer_expression
   | random_float_expression
   | random_boolean_expression ;
```

15. Error Handling

15.1 TRY-CATCH

```
   TRY
statements
   CATCH error
statements
   ENDTRY
```

The error identifier is optional.

```
   TRY
statements
   CATCH
statements
   ENDTRY
```

Formal rule:

```
   try_statement ::=
"TRY" ,
newline ,
{ statement | comment } ,
"CATCH" ,
[ identifier ] ,
newline ,
{ statement | comment } ,
"ENDTRY" ,
   newline ;
```

15.2 RAISE

Optional explicit error generation:

```
   RAISE expression
```

Example:

```
   RAISE STRING("Invalid input")
```

Formal rule:

```
   raise_statement ::=
"RAISE" ,
   expression ,
   newline ;
```

16. Class Rules

Class support is included but may be excluded from the first experimental subset if the 50 algorithms do

not require it.

16.1 Class Definition

```
CLASS ClassName
```

declarations

```
ENDCLASS
```

Formal rule:

```
class_definition ::=
```

"CLASS" ,

```
identifier ,
```

newline ,

{ constructor_definition | method_definition | property_declaration | comment } ,

"ENDCLASS" ,

```
newline ;
```

16.2 Constructor

The dialect uses THIS instead of Python's self.

```
CONSTRUCTOR ACCEPTS(x, y)
PROPERTY THIS.x <- x
PROPERTY THIS.y <- y
ENDCONSTRUCTOR
```

Formal rule:

```
constructor_definition ::=
```

"CONSTRUCTOR" ,

"ACCEPTS" ,

"(" ,

```
[ parameter_list ] ,
```

")" ,

newline ,

{ statement | property_assignment | comment } ,

"ENDCONSTRUCTOR" ,

```
newline ;
```

16.3 Property Declaration

```
PROPERTY name
```

or:

```
PROPERTY name AS INTEGER
```

Formal rule:

```
property_declaration ::=
"PROPERTY" ,
identifier ,
[ "AS" , type_name ] ,
newline ;
```

16.4 Property Assignment

```
PROPERTY THIS.name <- value
```

Formal rule:

```
property_assignment ::=
"PROPERTY" ,
"THIS" ,
"." ,
identifier ,
"<-" ,
expression ,
newline ;
```

16.5 Method Definition

```
METHOD name ACCEPTS(parameters)
statements
ENDMETHOD
```

Formal rule:

```
method_definition ::=
"METHOD" ,
identifier ,
"ACCEPTS" ,
"(" ,
[ parameter_list ] ,
")" ,
newline ,
```

{ statement | comment } ,

"ENDMETHOD" ,

```
   newline ;
```

16.6 Property Access

THIS.property_name

```
   object.property_name
```

Formal rule:

```
   property_access ::=
   identifier ,
"." ,
identifier ;
```

THIS is treated as a reserved contextual identifier inside classes.

16.7 Member-Method Invocation

v1.1 explicitly distinguishes property access from invocation by the presence of parentheses.

Canonical syntax:

```
   object.method(arguments)
```

Examples:

stack.pop()

drone.move(nextMove)

```
   registry.GET(name)
```

Formal rule:

```
   member_call_expression ::=
primary_expression ,
"." ,
   identifier ,
"(" ,
   [ argument_list ] ,
")" ;
```

The method identifier remains case-sensitive.

Parentheses distinguish a method invocation from property access:

```
x.item       COMMENT property access
x.item()     COMMENT zero-argument method invocation
```

A member call may be used as an expression or as a standalone statement.

17. Type Names

```
type_name ::=
```

"INTEGER"

```
| "FLOAT"
| "STRING"
| "CHAR"
| "BOOLEAN"
| "LIST"
| "DICTIONARY"
| "SET"
| "TUPLE"
| identifier ;
```

A user-defined class name may be used as a type.

18. Statement Grammar

```
statement ::=
```

assignment_statement

```
| output_statement
| call_statement
| append_statement
| insert_statement
| remove_first_statement
| remove_last_statement
| remove_value_statement
| if_statement
| while_statement
| range_for_statement
| for_each_statement
| return_statement
| break_statement
| continue_statement
| try_statement
| raise_statement ;
```

19. Declaration Grammar

```
   declaration ::=
```
function_definition
```
   | class_definition ;
```

20. Literal Grammar

```
   literal ::=
```
integer_literal
```
   | float_literal
   | typed_integer
   | typed_float
   | string_literal
   | char_literal
   | boolean_literal
   | none_literal
   | tuple_literal
   | typed_list_literal
   | mixed_list_literal
   | set_literal
   | empty_literal ;
```

21. Complete High-Level Grammar

```
   program ::=
```
{ declaration | statement | comment } ;

```
   declaration ::=
```
function_definition
```
   | class_definition ;
```

```
   statement ::=
```
assignment_statement
```
   | output_statement
   | call_statement
   | append_statement
   | insert_statement
   | remove_first_statement
   | remove_last_statement
   | remove_value_statement
   | if_statement
   | while_statement
   | range_for_statement
   | for_each_statement
   | return_statement
   | break_statement
   | continue_statement
   | try_statement
```

```
   | raise_statement ;

   expression ::=
```
or_expression ;

```
   or_expression ::=
```
xor_expression ,
{ "OR" , xor_expression } ;

```
   xor_expression ::=
```
and_expression ,
{ "XOR" , and_expression } ;

```
   and_expression ::=
```
not_expression ,
{ "AND" , not_expression } ;

```
   not_expression ::=
"NOT" , not_expression
   | comparison_expression ;

   comparison_expression ::=
```
additive_expression ,
[
   (
"=="
   | "!="
   | "<"
   | ">"
   | "<="
   | ">="
   | "IN"
   | "NOT IN"
),
additive_expression
];

```
   additive_expression ::=
```
multiplicative_expression ,
{ ("+" | "-") , multiplicative_expression } ;

```
   multiplicative_expression ::=
```
power_expression ,

```
{
   (
"*"
   | "/"
   | "INTEGER DIVISION"
   | "MODULUS"
),
power_expression
};

   power_expression ::=
unary_expression ,
[ "EXP" , power_expression ] ;

   unary_expression ::=
"-" , unary_expression
   | primary_expression ;

   primary_expression ::=
literal
   | identifier
   | indexed_access
   | property_access
   | member_call_expression
   | call_expression
   | input_expression
   | length_expression
   | join_expression
   | dictionary_get_expression
   | random_expression
   | set_operation
   | "(" , expression , ")" ;

   member_call_expression ::=
primary_expression ,
"." ,
   identifier ,
"(" ,
   [ argument_list ] ,
")" ;

   none_literal ::= "NONE" ;
```

22. Static Semantic Rules

The parser may produce a syntactically valid AST that still violates semantic rules. The following checks must therefore be performed after parsing.

22.1 Identifier Rules

An identifier must be declared or assigned before it is read.

Invalid:

```
OUTPUT total
total <- 0
```

22.2 Scope Rules

Variables declared within a function belong to that function.

Loop variables belong to the containing function or block, depending on the implementation policy.

Function parameters are available throughout the function body.

22.3 Type Compatibility

Operations must use compatible types.

Valid:

```
total <- INTEGER(5) + INTEGER(2)
```

Invalid:

```
total <- INTEGER(5) + STRING("two")
```

22.4 Boolean Conditions

Conditions used by IF and WHILE must evaluate to Boolean values.

22.5 Return Rules

A RETURN statement may only occur inside a function, method, or constructor where explicitly permitted.

Constructors should not return a value.

22.6 Break and Continue Rules

BREAK and CONTINUE may only occur inside loops.

22.7 Collection Rules

List indexes must be integers.

Dictionary keys must be hashable in the generated Python representation.

Collection operations must target compatible collection types.

22.8 Function Call Rules

The number of supplied arguments must match the number of required parameters unless optional parameters are later added.

22.9 Range Rules

The STEP value must not be zero.

Range loops are end-exclusive.

22.10 Character Rules

A CHAR value must contain exactly one character or one valid escape sequence.

22.11 Member Calls and Dictionary GET Resolution

Property access object.property and member invocation object.method(...) are distinct constructs.

A member call is resolved according to the receiver type or other static evidence.

When the receiver is dictionary-backed and the method identifier is GET, the call resolves to the dictionary intrinsic.

The dictionary intrinsic GET accepts exactly one or two arguments.

When the receiver is not dictionary-backed, GET remains an ordinary user-defined method identifier.

An unknown receiver type must not be assumed to be a dictionary merely because the method is named GET.

NONE is a literal value and is exempt from identifier declaration-before-use checks.

23. Normalisation Rules

The evaluation framework may perform conservative normalisation before strict parsing.

Permitted transformations include:

Keyword case normalisation

Unicode arrow → <-

Removal of Markdown code fences

Line-ending normalisation

Trailing whitespace removal

Approved spelling correction

Approved synonym mapping

v1.1 compatibility normalisation may also include:

```
Legacy dictionary~GET(...) -> dictionary.GET(...)
```

Case normalisation of None / none -> NONE when used as a standalone null literal

These transformations must obey the same logging and semantic-preservation requirements as all other normalisation.

Examples:

while x < 5 do

may become:

```
WHILE x < 5 DO
```

x←5

may become:

```
x <- 5
```

The framework must log every transformation.

It must not:

invent missing variables;

insert missing algorithmic operations;

change conditions;

add missing branches;

alter arithmetic operators;

repair semantic logic silently.

24. Guided Link Rules

Guided Links are documentation annotations and do not alter program execution.

Recommended syntax:

```
GUIDED LINK("typed lists")
```

Formal rule:

```
guided_link ::=
```

"GUIDED LINK" ,

"(" ,

quoted_string ,

")" ,

```
newline ;
```

Guided Links may appear between statements and should be:

- stored as metadata;
- excluded from AST execution nodes;
- excluded from generated Python;
- optionally shown in teaching-oriented visualisations.

25. Complete Example

```
FUNCTION find_largest ACCEPTS(numbers AS LIST)
IF LENGTH OF numbers == 0 THEN
RAISE STRING("The list cannot be empty")
ENDIF

largest <- numbers[0]
```

```
  FOR EACH number IN numbers DO
  IF number > largest THEN
  largest <- number
  ENDIF
  ENDFOR

  OUTPUT STRING("The largest value is {largest}")
  RETURN largest
  ENDFUNCTION
```

26. Expected AST Outline

ProgramNode

└── FunctionNode: find_largest

├── ParameterNode: numbers AS LIST

└── Body

├── IfNode

│ ├── Condition

│ │ └── EqualNode

│ │ ├── LengthNode

│ │ │ └── IdentifierNode: numbers

│ │ └── IntegerLiteralNode: 0

│ └── ThenBody

│ └── RaiseNode

│ └── StringLiteralNode

├── AssignmentNode

│ ├── IdentifierNode: largest

│ └── IndexAccessNode

│ ├── IdentifierNode: numbers

│ └── IntegerLiteralNode: 0

├── ForEachNode

│ ├── Iterator: number

│ ├── Collection: numbers

│ └── Body

│ └── IfNode

│ ├── GreaterThanNode

│ └── AssignmentNode

├── OutputNode

└── ReturnNode

27. Versioning Rule

This specification is frozen as:

Cambridge-Style Pseudocode - AJ_24038155 Dialect v1.1

v1.1 supersedes the v1.0 specification by integrating the following approved language changes:

NONE as a native null literal;

general member-method invocation using object.method(...);

```
canonical dictionary retrieval using dictionary.GET(...);
backwards-compatible acceptance of the v1.0 dictionary~GET(...) form.
```

Any later syntax addition, removal, or semantic alteration must create a new version, for example:

AJ_24038155 Dialect v1.2

AJ_24038155 Dialect v2.0

AI outputs should always be evaluated against the exact grammar version recorded at generation and evaluation time.